TICKET 7

2024-2025 - 12.06

1. Regular expression of the grammar:

S -> aS | aA
A -> bA | b

a) a*b*
b) a*b+
c) (ab)*
d) (ab)+

Formula:

$X = aX + b$

$X = a^* b$

$S \to aS + a \to aA$

$A \to bA + b$

$A = b^* b = b^+$

$S = aS + a + ab^+$

$S = a^*(a + ab^+)$

$S = a^+ + a^+ b^+ = a^+ b^*$

2. Words that belong to the grammar:

S -> 0A
A -> 1S | AA | 0

a) 00
b) 100
c) 010
d) 0100

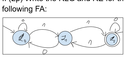
3. L = { a, aa }. ? ∈ L*

a) ε
b) a
c) aaaa
d) aaa

4. L_1 = { ε, a }, L_2 = { b, c }. $L_1 \oplus L_2$ = ?

a) ε, a, b
b) ab, bc
c) a
d) a, b, c

II (2p) Write the RLG and RE for the following FA:



$q_0 = q_0 0 + q_1 0 + \varepsilon$

$q_1 = q_0 1 + q_2 1$

$q_2 = q_2 0 + q_1 1$

$X = Xa + b$

$X = ba^*$

$q_2 = q_1 1 0^*$

$q_1 = q_0 1 + q_1 1 0^* 1$

$q_1 = q_0 1 (1 0^* 1)^*$

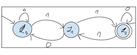
$q_0 = q_0 0 + q_0 1 (10^* 1)^* 0 + \varepsilon$

$q_0 = q_0 (0 + 1(10^* 1)^* 0) + \varepsilon$

$q_0 = \varepsilon (0 + 1(10^* 1)^* 0)^* = 0 + 1(10^* 1)^* 0)^*$

$q_1 = (0 + 1(10^* 1)^* 0)^*) 1 (10^* 1)^*$

$q_2 = (0 + 1(10^* 1)^* 0)^*) 1 (10^* 1)^* 1 0^*$



$G = \{N, \Sigma, P, S\}$

$N = \{q_0, q_1, q_2\}$

$S = q_0$

$\Sigma = \{0, 1\}$

P: $\delta(q_0, 0) = q_0$ $\quad q_0 \to 0 q_0$

$\delta(q_0, 1) = q_1$ $\quad q_0 \to 1 q_1$

$\delta(q_1, 1) = q_2$ $\quad q_1 \to 1 q_2$

$\delta(q_1, 0) = q_0$ $\quad q_1 \to 0 q_0$

$\delta(q_2, 0) = q_2$ $\quad q_2 \to 0 q_2$

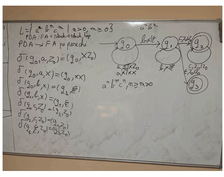
$\delta(q_2, 1) = q_1$ $\quad q_2 \to 1 q_1$

$q_0 \to 0 q_0 \mid 1 q_1$

$q_1 \to 1 q_2 \mid 0 q_0 \mid 1$

$q_2 \to 0 q_2 \mid 1 q_1 \mid 0$

III (1p) PDA for L = { $a^n b^m c^m$ | n > 0, m >= 0 }



IV (2p)

G = (E, A (;) ; P, S)

P: E -> (E) | a

A -> A;E | E

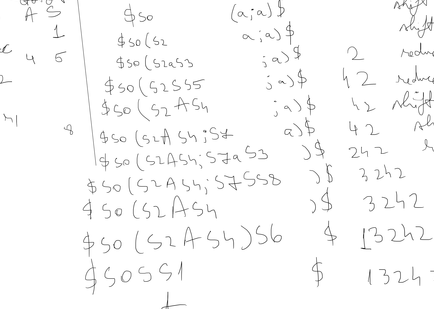
LR(0) of canonical collection of states is:

$s0 = \{$

$s1 = \ldots$

and FOLLOW(E) = { $,) } and FOLLOW(A) = { ; }.

Construct the SLR table and parse the sequence w = (a;a)



	Action					Goto	
	(	)	;	a	$	A	S
0	2					1	
1					acc		
2				3		4	5
3	r3	r3	r3				
4			6				
5			r4	r4			
6			r1	r1		7	
7	2					8	
8			r2	r2			

Stack	Input	Output	Action
$s0	(a;a)$		shift
$s0(s2	a;a)$		shift
$s0(s2a s3	;a)$	2	reduce
$s0(s2S s5	;a)$	r2	reduce
$s0(s2A s4	;a)$	r2	shift
$s0(s2A s4;s6	a)$	r2	shift
$s0(s2A s4;s6a s3	)$	r2 r2	reduce reduce
$s0(s2A s4;s6S s8	)$	r3 r2	
$s0(s2A s4	)$	r3 r2	shift
$s0(s2A s4)s6	$	r3 r2 r2	reduce
$s0S s1	$	r3 r2 r2	accept
accept			

V (2p)

Define a grammar corresponding to arithmetic expressions (with operators for addition and subtraction) and then define an attribute grammar for counting the number of operators and number of operands.

$G = \{\{S, OP\}, \{+, -, const\}, P, S\}$

$AG = \{G, \{n, m\}, R\}$

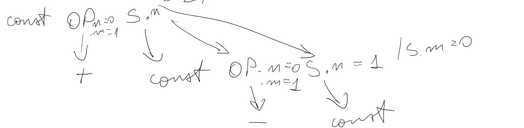
$S \to const$

$S \to const\ OP\ S$

$OP \to + \mid -$

$S.n = 1 \quad S.m = 0$

$S_1.m = 1 + OP.n + S_2.n \quad S_1.m = OP.m + S_2.m$

$OP.n = 0 \quad OP.m = 1$

$2 + 2 - 1$